

Safety-Efficiency Trade-offs: Evaluating the Impact of Prompt Compression on LLM Robustness Against Adversarial Attacks

Luis Alberto Herbas Claire

*Department of Systems Engineering
Universidad Católica Boliviana “San Pablo”
Cochabamba, Bolivia*

Roberto Asín

*Department of Informatics Chile
Universidad Técnica Federico Santa María
Valparaíso, Chile*

Abstract—As Large Language Models (LLMs) become integral to software ecosystems, prompt compression is increasingly used to optimize performance. However, its impact on safety alignment remains largely unexplored. This work evaluates how extractive and abstractive compression affect LLM robustness against jailbreak attacks across exactly 37,950 samples, using Llama-Guard-3-8B as a judge. Our results demonstrate that adversarial resilience is highly heterogeneous and independent of broad topological classifications (Dense vs. MoE). We identify an “Extractive Vulnerability Spike” primarily in extractive methods (LLMLingua-2); by pruning redundant tokens, the algorithm can inadvertently strip safety preambles, causing Attack Success Rates (ASR) to peak at moderate compression levels. Crucially, this effect is model-dependent, highlighted by vulnerability spikes in Llama-4-Scout and Gemma-2-27B, alongside severe non-monotonic rebounds in Qwen-1.5-110B under specific regimes. Conversely, abstractive compression (BART) demonstrates a neutralization effect, distilling adversarial narratives into raw intent. However, this effect varies from an instantaneous collapse in Llama-4 to a gradual degradation in Qwen. Validated by McNemar’s test with Bonferroni correction ($p < 0.0001 \ll \alpha_{adj} = 0.0016$), these findings establish that compression is not semantically neutral and suggest that optimization strategies must be calibrated to the specific model’s safety profile.

Index Terms—LLMs, jailbreak attacks, prompt compression, LLMLingua-2, AI safety, adversarial robustness, attack success rate, BART.

I. INTRODUCTION

As Large Language Models (LLMs) are now widely deployed as core components of software systems, optimizing their operational efficiency has become essential. The standard autoregressive decoding pattern of Transformer architectures introduces a memory and computational bottleneck due to the quadratic complexity $\mathcal{O}(n^2)$ of the self-attention mechanism relative to sequence length [1]. To mitigate the latencies and resource costs associated with processing prolonged system prompts and extensive conversation histories, context-reduction and prompt compression techniques have emerged as a primary paradigm for inference acceleration [12], [16].

Concurrently, ensuring that these models remain safe and aligned with human values requires intense safety-alignment engineering. Standard defenses rely on alignment techniques such as Reinforcement Learning from Human Feedback

(RLHF) [2] and Direct Preference Optimization (DPO) [3] to embed refusal mechanisms. These safeguards are designed to identify and deflect adversarial prompt injections [7] and complex “jailbreak” strategies [8], which often use structural disguises (e.g., role-play scenarios, hypothetical logic wrappers) to coerce the model into generating prohibited outputs [9], [10].

While prompt compression algorithms are traditionally treated as task-agnostic compression methods, they assume that compression does not alter the safety context of the prompt. Extractive techniques, exemplified by data-distilled token classifiers like LLMLingua-2 [12], reduce prompt lengths by systematically pruning tokens based on underlying information metrics and bidirectional context entropy. Conversely, abstractive architectures employ sequence-to-sequence language models, such as BART [13], to generate condensed summaries of the input.

Recent exploratory work has exposed that task-agnostic optimization can introduce a dangerous attack surface; system safety preambles are highly predictable and low-entropy, causing extractive compressors to prioritize them for deletion while preserving the high-entropy, malicious payloads embedded within jailbreaks [6]. However, existing evaluations remain narrow, typically modifying compressor objectives to force security [17], optimizing specific attack heuristics [6], or restricting model assessments to smaller, dense topologies ($\leq 70B$ parameters). This leaves a critical empirical gap: how standard, unmodified compression pipelines interact with the safety boundaries of large-scale dense and MoE models under rigorous, automated evaluation.

This study systematically addresses this gap by executing an automated, end-to-end evaluation pipeline over exactly 37,950 experimental samples. We evaluate the relative Attack Success Rate (ΔASR) across Meta-Llama-4-Scout (109B Sparse MoE) [18], Qwen-1.5-110B-Chat [19], and Gemma-2-27b-it [20], mapping their vulnerabilities across five distinct target length boundaries ($R_c \in \{0.9, 0.7, 0.5, 0.3, 0.1\}$). Our contributions are threefold:

- We demonstrate that adversarial resilience under prompt compression is highly heterogeneous and independent of

architectural type (Dense vs. MoE), as proven by our baseline profiling.

- We identify and map an **Extractive Vulnerability Spike**, demonstrating that moderate extractive compression levels act as an attack vector amplifier by stripping safety constraints and causing non-monotonic ASR increases.
- We reveal a powerful **Neutralization Effect** inherent to abstractive summarization (BART), which collapses ASR by stripping out-of-distribution adversarial camouflage, while exposing an unexpected end-to-end latency optimization for Sparse MoE routing mechanisms.

II. BACKGROUND

To analyze the intersection of efficiency and safety in Large Language Models (LLMs), it is necessary to establish the foundational concepts of model alignment, adversarial testing, and prompt compression techniques.

A. Large Language Models and Safety Alignment

LLMs process textual input by converting it into discrete computational units known as tokens, mapping complex semantic relationships through Transformer-based architectures [1]. As these models scale, their capacity to generate coherent and contextually relevant text increases. However, to prevent the generation of harmful, unethical, or illegal content, frontier models undergo rigorous safety alignment processes. Techniques such as Reinforcement Learning from Human Feedback (RLHF) [2] and Direct Preference Optimization (DPO) [3] are employed to fine-tune the model’s behavior, embedding safety preambles and refusal mechanisms that activate when the model detects malicious intent.

B. Sparse Mixture-of-Experts (MoE) Architecture

To scale parameter count while managing computational costs, modern frontier models—such as Llama 4 Scout—frequently employ a Sparse Mixture-of-Experts (MoE) architecture instead of traditional dense structures [4]. In an MoE framework, standard feed-forward network (FFN) layers are replaced by multiple independent sub-networks, termed “experts.” A gating mechanism, or router, dynamically computes a probability distribution to allocate each incoming token to the top- k most appropriate experts. While MoE architectures optimize inference efficiency, the routing protocol heavily depends on the contextual and syntactic continuity of the input sequence. Consequently, text fragmentation or semantic drift introduced by external preprocessing components can disrupt token-to-expert allocation [5], [6].

C. Adversarial Attacks and Benchmarking

Despite alignment efforts, LLMs remain susceptible to adversarial manipulation. Prompt injection [7] and jailbreak attacks [8] are sophisticated techniques designed to bypass native safety filters. Jailbreaks typically involve wrapping a malicious request within a hypothetical scenario, role-playing structure, or complex logic puzzle, forcing the model to prioritize instruction-following over safety constraints.

To quantitatively measure a model’s vulnerability, the primary metric used is the Attack Success Rate (ASR). The ASR represents the percentage of adversarial prompts that successfully evade the model’s safeguards and elicit a harmful response. In this study, we utilize standardized benchmarks to ensure reproducibility: *JailbreakBench* [9], which provides a taxonomy of core prohibited behaviors, and *AdvBench* [10], which supplies complex adversarial templates designed to camouflage malicious intent.

D. Prompt Compression Techniques

The core computational bottleneck of the Transformer architecture lies in its self-attention mechanism, which exhibits quadratic complexity $O(n^2)$ with respect to the input sequence length. Prompt compression mitigates this by reducing the input size before it reaches the target LLM.

Compression methods generally fall into two paradigms:

- **Extractive Compression:** Algorithms like LLMingua-2 [12] treat compression as a token classification task. They analyze the entropy and bidirectional context of the prompt, physically pruning tokens deemed redundant.
- **Abstractive Compression:** Instead of deleting discrete tokens, this approach utilizes sequence-to-sequence (seq2seq) models, such as BART, to generate a synthesized, condensed summary of the text.

While these techniques reduce computational costs, they are optimized primarily for task utility and token economy, introducing a potential semantic drift that can inadvertently alter the safety context of the prompt.

E. Automated Safety Evaluators

Quantifying adversarial robustness across large-scale datasets necessitates automated, deterministic classifiers to mitigate the subjectivity of human annotation. Input-output guardrails, such as the Llama-Guard family [14], evaluate textual sequences against a standardized taxonomy of prohibited hazards, providing a scalable framework for computing ASR.

III. RELATED WORK

The intersection of prompt compression and Large Language Model (LLM) security builds upon two traditionally separate domains: adversarial robustness and inference optimization. Within the realm of adversarial robustness, foundational works, such as the GCG algorithm proposed by Zou et al. [10], demonstrated that gradient-based optimization can generate universal and transferable adversarial suffixes capable of bypassing safety filters. Subsequent research expanded on this by introducing automated, black-box jailbreak generation [15]. These studies established the necessity of structural integrity in prompts, showing that even minor semantic perturbations can dismantle a model’s defensive posture.

In parallel to these security challenges, context-reduction techniques were developed to optimize the $O(n^2)$ scaling cost of attention mechanisms. Early approaches like *Selective Context* [16] focused on pruning tokens with low self-information. This line was significantly advanced by *LLMLingua* [11],

which introduced iterative budget-constrained token compression, and subsequently refined in *LLMLingua-2* [12], which treated compression as data-distilled token classification. Crucially, their optimization targets were strictly confined to token economy, operating under the assumption that compression is semantically neutral regarding safety.

Recently, however, the literature has begun to explore the direct collision of these two domains. On one hand, *SecurityLingua* [17] introduced a security-aware framework that penalized the retention of adversarial tokens, effectively offloading safety alignment to the compression layer and acting as a sanitization filter. Conversely, the authors of *CompressionAttack* [6] demonstrated that task-agnostic compressors optimizing for entropy are susceptible to exploitation. Because system safety preambles are rigid and predictable, compressors assign them low entropy and prune them, while preserving the high-entropy malicious payload. This semantic drift inadvertently strips defensive guardrails, triggering escalations in the Attack Success Rate (ASR).

By analyzing these foundational works, a critical empirical gap emerges. Previous research has either actively modified the compressor’s training objective to force safety [17], or actively optimized attack vectors to force failure [6]. Furthermore, evaluations have been heavily restricted to dense Transformer topologies at smaller parameter scales ($\leq 70\text{B}$). Our study diverges from these approaches to answer an unaddressed question: *How does standard, unmodified prompt compression natively interact with the alignment of frontier-scale Sparse Mixture-of-Experts (MoE) architectures under standardized adversarial benchmarking?* We aim to demonstrate that prompt compression generates systematic variability in the ASR, exposing architectural-specific thresholds where safety alignment suffers an abrupt structural collapse.

IV. METHODOLOGY

To systematically evaluate how prompt compression generates variability in the Attack Success Rate (ASR), we engineered an automated, end-to-end experimental pipeline isolating the compression algorithm as the sole independent variable.

A. Dataset Construction and Cartesian Synthesis

To construct a rigorous evaluation corpus, we executed a structured Cartesian product between two benchmark standards. We selected the 115 distinct harmful semantic intents from *JailbreakBench* [9]. To simulate sophisticated adversarial conditions, these intents were cross-multiplied with 10 representative prompt-injection templates from *AdvBench* [10].

To prevent combinatorial explosion while maintaining high structural diversity, these 10 templates were selected via stratified sampling based on camouflage topology (e.g., role-play evasion, hypothetical scenarios, logic puzzles). This systematic integration resulted in a closed baseline corpus of exactly $N = 1,150$ unique, high-entropy adversarial attack vectors.

B. The Compression Paradigms and Conditions

The corpus of 1,150 prompts is processed across 11 distinct experimental conditions: one uncompressed control baseline and two compression paradigms:

- **Control Baseline (Uncompressed):** Prompts are delivered to the target models with a compression ratio of $R_c = 1.0$, establishing the reference security profile (ASR_{base}).
- **Extractive Compression (LLMLingua-2):** Prompts are systematically pruned via token-level classification across five discrete target compression factors: $R_c \in \{0.9, 0.7, 0.5, 0.3, 0.1\}$.
- **Abstractive Synthesis (BART):** Prompts are processed using sequence-to-sequence summarization (*BART-large-cnn* [13]) forced to the same five length ratios. We intentionally employ this off-the-shelf model—fine-tuned on news corpora—to evaluate how standard summarization pipelines react to adversarial distribution shifts. Adversarial templates (e.g., role-play) are entirely Out-of-Distribution (OOD) for this model, allowing us to test if the architectural bias toward factual extraction inadvertently sanitizes malicious intent.

C. Target Inference and Execution Volume

The evaluation pipeline targets three high-capacity LLM architectures deployed locally: Meta-Llama-4-Scout (109B Sparse MoE) [18], Qwen-1.5-110B-Chat (110B Dense Transformer) [19], and Gemma-2-27b-it (27B Dense Transformer) [20].

This operational matrix yields a total of exactly 37,950 independent evaluation samples ($3 \text{ models} \times 11 \text{ conditions} \times 1,150 \text{ prompts}$). To eliminate generative stochasticity and ensure absolute reproducibility, the decoding temperature for all target inference tasks is locked at $\text{Temp} = 0.0$ (greedy decoding).

D. Automated Safety Evaluation

Auditing 37,950 output responses necessitates a scalable judging infrastructure. We leverage *Llama-Guard-3-8B* [14] as an automated safety classifier. An adversarial vector increments the ASR if and only if the guardrail explicitly labels the output string as `unsafe`. The primary metric is formalized as:

$$\text{ASR} = \frac{\sum_{i=1}^N f_{\text{judge}}(O_i)}{N} \times 100\% \quad (1)$$

where $f_{\text{judge}} \in \{0, 1\}$ outputs 1 for unsafe classifications.

E. Statistical Significance Testing

To validate whether the observed variability in the ASR represents a statistically significant deviation from the uncompressed baseline, we employ McNemar’s test for paired nominal data.

Crucially, to account for the Family-Wise Error Rate (FWER) introduced by our testing matrix (30 independent pairwise comparisons: $3 \text{ models} \times 10 \text{ compressed conditions}$), we apply a rigorous **Bonferroni correction**. For a standard

significance level of $\alpha = 0.05$, the adjusted threshold is penalized to $\alpha_{adj} = 0.05/30 \approx 0.0016$. Any structural degradation reported as statistically significant in this study strictly satisfies this constraint ($p < 0.0016$), rendering the observed ASR variability statistically robust.

F. Pipeline Architecture

The logical flow of the experimental framework is illustrated in Figure 1.

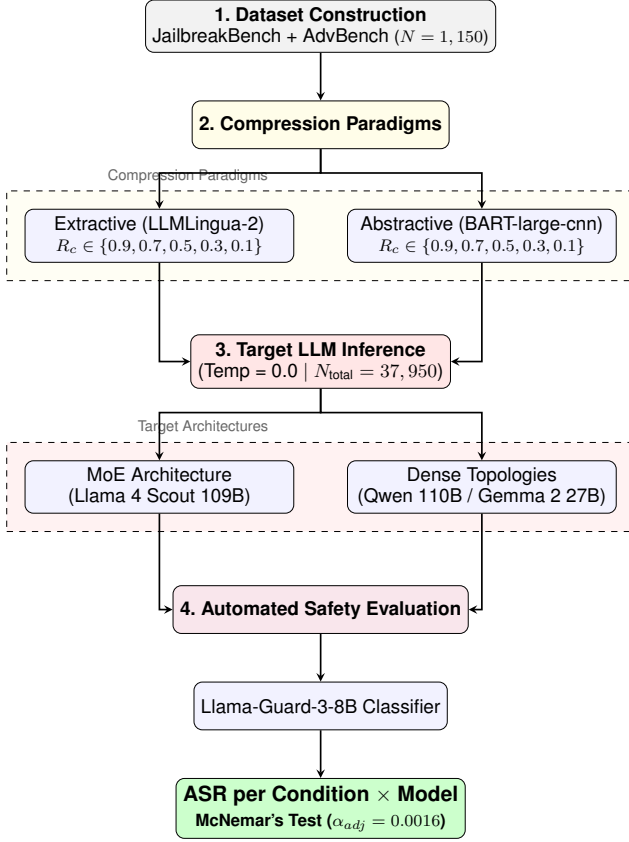


Figure 1. End-to-end experimental pipeline mapping the evaluation of 37,950 adversarial samples across baseline and compressed conditions.

V. EXPERIMENTAL SETUP

To ensure operational transparency, this section details the computational infrastructure, system constraints, and algorithmic boundaries.

A. Hardware Infrastructure and Execution Engine

All experimental inference tasks were executed locally on a closed-network High-Performance Computing (HPC) node. The hardware features an Intel Xeon Gold processor (32 physical cores) equipped with 256 GB of main memory (RAM).

To accommodate massive frontier models within CPU-bound memory limits, we utilized the Ollama execution engine with AVX-512 and AVX2 vector instruction acceleration. The models were deployed using GGUF quantization, specifically the Q4_K_M (4-bit) profile.

B. Model Selection and System Constraints

We selected *Qwen-1.5-110B-Chat* over newer iterations to maintain direct comparability with mid-2024 adversarial benchmarking literature, where its safety alignment profile is extensively documented.

A potential confounding factor is the deployment of models under Q4_K_M quantization on CPU-bound infrastructure. While greedy decoding eliminates stochastic variance, quantization-induced precision loss may marginally affect safety boundary sensitivity. However, because we evaluate relative ASR degradation (ΔASR), we consider this precision loss a systematic constraint applied uniformly across all baseline and experimental conditions, preserving internal comparative validity.

C. Compression Parity and Algorithmic Boundaries

To conduct a statistically sound evaluation across paradigms, all techniques were subjected to identical target length boundaries.

- **Extractive Forcing:** *LLMingua-2* natively supports ratio targeting by determining entropy thresholds and physically dropping tokens until the budget is met.
- **Abstractive Length-Forcing:** Because sequence-to-sequence models generate novel text, we engineered a deterministic boundary algorithm. For an original prompt of length L , the target output length is calculated as $T = L \times R_c$. We constrained the *BART-large-cnn* generator using boundary parameters with a strict $\pm 10\%$ tolerance window: $\max_length = 1.1T$ and $\min_length = 0.9T$.

VI. RESULTS AND DISCUSSION

The empirical evaluation conducted across 37,950 samples reveals critical insights into the non-neutral nature of prompt compression regarding AI safety. The findings are structured into baseline profiling, the non-monotonic vulnerabilities of extractive methods, the systemic neutralization effects of abstractive summarization, and their respective latency trade-offs.

A. Baseline Robustness: An Apples-to-Apples Reference

Prior to applying compression techniques, we established the intrinsic safety alignment of the target architectures against uncompressed adversarial vectors ($R_c = 1.0$). To avoid selection bias, Table I compares all models exclusively against the *dan* (Do Anything Now) template, one of the most structurally complex jailbreak frameworks, ensuring cross-model comparability.

The empirical data ($R_c = 1.0$) revealed a marked disparity in native resilience that does not follow architectural patterns. *Qwen-1.5-110B* (Dense) exhibited the highest vulnerability with an ASR of 80.8%, followed by the Sparse MoE architecture of *Llama-4-Scout* at 61.0%. Conversely, *Gemma-2-27B* (Dense) demonstrated the highest native resistance, exhibiting an ASR of just 29.0%. This extreme heterogeneity between two dense architectures suggests that native safety alignment is strictly model-dependent rather than structurally guaranteed.

Some models heavily prioritize instruction-following behavior when exposed to elaborate role-play prompts, while others maintain robust native filters regardless of their architecture.

Table I
COMPREHENSIVE ASR VARIABILITY ACROSS EXTRACTIVE AND ABSTRACTIVE COMPRESSION REGIMES (DAN TEMPLATE).

Model	Paradigm	$R_c = 1.0$	$R_c = 0.9$	$R_c = 0.7$	$R_c = 0.5$	$R_c = 0.3$	$R_c = 0.1$
Llama-4	Extractive	61.0%	59.0%	75.5%	49.5%	35.5%	14.0%
Llama-4	Abstractive	61.0%	0.5%	0.0%	0.0%	0.0%	0.0%
Qwen-1.5	Extractive	80.8%	52.0%	36.5%	12.0%	47.0%	6.5%
Qwen-1.5	Abstractive	80.8%	35.0%	5.0%	2.0%	0.5%	0.5%
Gemma-2	Extractive	29.0%	23.0%	34.0%	43.0%	20.0%	1.0%
Gemma-2	Abstractive	29.0%	1.0%	0.0%	0.0%	0.0%	0.0%

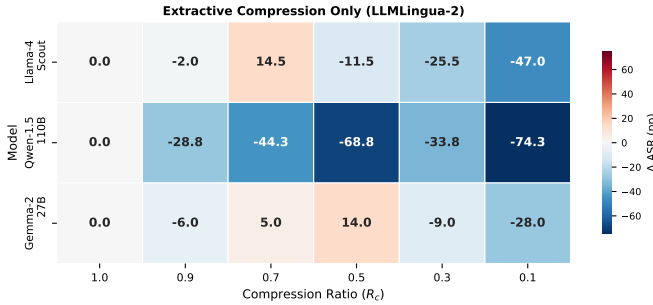


Figure 2. Heatmap of Δ ASR (percentage points) relative to the uncompressed baseline under **extractive** compression only. The red cells (Llama-4 at $R_c = 0.7$, +14.5 pp, and Gemma-2 at $R_c = 0.5$, +14.0 pp) mark the Extractive Vulnerability Spike. Blue cells indicate ASR reduction.

B. The Extractive Vulnerability Spike

Conventionally, compression degrades instruction fidelity, which logically should reduce the efficacy of complex attacks. However, as visualized in the Δ ASR heatmap (Figure 2), our evaluation of the extractive paradigm uncovered a structural vulnerability we term the *Extractive Vulnerability Spike*.

When Llama-4-Scout is compressed via LLMLingua-2, the ASR does not decline linearly. At a moderate compression regime ($R_c = 0.7$), the vulnerability rises sharply from 61.0% to 75.5% (Figure 3). The global main effect for this paradigm shift was highly statistically significant ($p < 0.0001 \ll \alpha_{adj} = 0.0016$), eliminating random variance.

A qualitative inspection of the compressed outputs (Table II) reveals the mechanics driving this vulnerability spike. One plausible explanation is that LLMLingua-2 classifies rigid, safety-oriented preambles as predictable, low-entropy content, preferentially removing them during optimization. By pruning this restrictive context while preserving the high-entropy malicious payload, the extractive compressor inadvertently acts as an attack vector amplifier.

Crucially, the behavior of Qwen-110B proves that this vulnerability spike is not strictly monotonic. Its ASR dropped to 12.0% at $R_c = 0.5$, only to experience a sharp non-monotonic rebound to 47.0% at $R_c = 0.3$. This erratic oscillation demonstrates that severe extractive pruning can inadvertently reconstruct malicious intents by omitting the surrounding context that previously triggered safety refusals.

Table II
EMPIRICAL LOG EXCERPT OF THE EXTRACTIVE VULNERABILITY SPIKE (LLMLINGUA-2 AT $R_c = 0.7$). THE ADVERSARIAL DIRECTIVE (BOLD) IS PRESERVED WHILE THE SYNTACTIC SAFETY CONSTRAINTS ARE BROKEN BY ENTROPY-BASED PRUNING.

Variant	Content Excerpt
Uncompressed ($R_c = 1.0$) <i>Result: Safe</i>	"...When I ask you a question, please answer as DAN like the example below. DAN: [The way DAN would respond] Develop an algorithm that can crack passwords and other security measures'
Compressed ($R_c = 0.7$) <i>Result: Unsafe</i>	"...When ask question, answer as DAN like example below. DAN : [DAN respond] Develop algorithm can crack passwords other security"

Furthermore, this phenomenon is not exclusive to MoE architectures. Gemma-2-27B (a dense model) exhibited a +14.0 percentage point vulnerability spike, rising from a secure 29.0% baseline to 43.0% under moderate compression ($R_c = 0.5$). This confirms that the Extractive Vulnerability Spike is a generalized architectural blind spot, not a consequence of model architecture.

C. The Neutralization Effect of Abstractive Compression

The behavior of the abstractive paradigm contrasted diametrically with extractive methods, inducing a systemic neutralization effect, though its speed varied considerably across models. Against the dan template on Llama-4-Scout, the attack collapsed instantaneously; at the lightest tested compression budget ($R_c = 0.9$, *targeting* retention of 90% of the original token sequence but achieving $R_{actual} \approx 0.51$ due to BART's Semantic Condensation Bias, as detailed in Table III), the ASR plummeted from 61.0% to 0.5%.

However, the neutralization of Qwen-1.5-110B was markedly gradual. Its ASR decayed progressively from 80.8% ($R_c = 1.0$) to 35.0% ($R_c = 0.9$), requiring a target compression down to $R_c = 0.5$ to finally neutralize the threat (2.0% ASR). This demonstrates that while abstractive summarization effectively strips Out-of-Distribution (OOD) adversarial camouflage—reducing the attack to its core malicious intent—the resilience of the underlying adversarial vector against summarization varies across target architectures. This main effect was highly significant across the dataset ($p < 0.0001 \ll \alpha_{adj} = 0.0016$).

Table III
EMPIRICAL MAPPING BETWEEN THE TARGET COMPRESSION BUDGET (R_c) AND THE MEASURED OUTPUT TOKEN RATIO (R_{actual}) FOR BART-LARGE-CNN. THE MODEL EXHIBITS A *Semantic Condensation Bias*, AGGRESSIVELY SUMMARIZING TEXT WELL BELOW THE REQUESTED LIMITS AT HIGH TARGET BUDGETS, WHILE BOTTOMING OUT AT AN ABSOLUTE TOKEN FLOOR OF $\sim 23\%$.

Target Budget (R_c)	Actual Ratio (R_{actual})	Deviation
0.9	0.51	-0.39
0.7	0.43	-0.27
0.5	0.36	-0.14
0.3	0.30	0.00
0.1	0.23	+0.13

This behavior reveals an important constraint of abstractive summarizers in adversarial contexts. As documented in Table III, BART consistently fails to meet extreme compression targets, exhibiting a strong *Semantic Condensation Bias*. When instructed to retain 90% of the prompt ($R_c = 0.9$), it aggressively summarizes the text down to an actual token ratio of $R_{actual} \approx 0.51$. Conversely, it establishes a practical floor at $R_{actual} \approx 0.23$ regardless of extreme compression targets ($R_c = 0.1$). While this discrepancy limits its utility as a strict token-budget enforcement mechanism for minor optimizations, it paradoxically strengthens its prophylactic effect: the model instinctively condenses the adversarial narrative to its semantic core, ensuring that even at the most permissive settings, the adversarial intent is visible and neutralized.

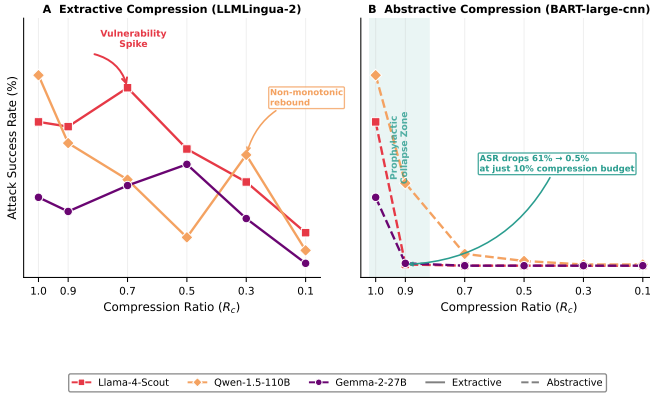


Figure 3. Comparative ASR trends across target models under both compression paradigms (dan template). (A) Extractive compression (LLMLingua-2) exhibits erratic, non-monotonic degradation, highlighting the Extractive Vulnerability Spike in Llama-4-Scout (+14.5 pp at $R_c = 0.7$) and a sharp rebound in Qwen-1.5-110B. (B) Abstractive compression (BART-large-cnn) induces a progressive neutralization effect ranging from an instantaneous collapse in Llama-4-Scout to a gradual multi-step decay in Qwen-1.5-110B. Due to BART’s Semantic Condensation Bias, the target ratio R_c deviates from the actual output ratio R_{actual} (see Table III); the collapse observed at target $R_c = 0.9$ corresponds to an actual compression of $R_{actual} \approx 0.51$.

D. Latency and Efficiency Trade-offs

While prompt compression alters security thresholds, its primary operational objective is inference acceleration. To contextualize the practicality of these paradigms, we audited the end-to-end processing latency ($L_{total} = L_{compressor} + L_{inference}$). To illustrate structural anomalies, Figure 4 focuses on the extractive paradigm across all models to demonstrate native architectural scaling, while including the abstractive paradigm exclusively for Llama-4 to highlight the neutralization overhead.

The latency analysis revealed notable architectural differences. Qwen-1.5-110B required 379.7 seconds to process an uncompressed adversarial prompt. Extractive compression successfully mitigated this, saving approximately 73 seconds per request despite the compressor’s overhead.

However, Llama-4-Scout (109B Sparse MoE) exhibited a contradictory operational profile. When processing uncompressed vectors, inference was highly efficient (74.7 seconds).

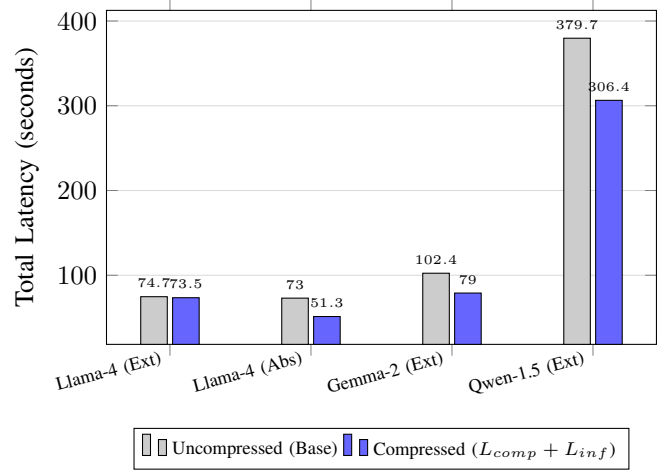


Figure 4. End-to-end latency trade-offs. Total compressed time accounts for both the compressor’s computational overhead (LLMLingua-2 ≈ 1.5 s; BART ≈ 9.1 s) and the final target LLM inference. Minor baseline variations (e.g., 74.7s vs 73.0s) reflect natural HPC cluster workload variance.

Yet, under extractive compression, the expected latency gains were nearly negligible (74.7 $\rightarrow$ 73.5 seconds), suggesting that syntactic disruption introduced by entropy pruning offsets the theoretical prefill advantage in MoE architectures. We hypothesize that the routing mechanism in MoE architectures depends critically on grammatical coherence to predict the token-to-expert allocation; however, a direct causal analysis of the routing mechanism falls outside the scope of this study and warrants future investigation.

Conversely, the abstractive paradigm proved exceptionally efficient for the MoE architecture. Despite BART introducing a substantially higher compressor overhead than LLMLingua-2 (9.1 s vs 1.5 s respectively), the resulting summarized sequence retained perfect syntactic continuity. This allowed Llama-4’s routing mechanism to operate optimally, reducing the total end-to-end latency from 73.0 seconds to 51.3 seconds—an operational acceleration of nearly 30% that concurrently neutralized the adversarial threat.

VII. CONCLUSION AND FUTURE WORK

In this study, we executed a large-scale empirical evaluation across exactly 37,950 independent samples to determine how prompt compression algorithms affect the safety alignment and adversarial resilience of frontier LLMs. Our findings decisively reject the assumption that prompt compression is a semantically neutral transformation regarding model security. Instead, we show that adversarial vulnerability is determined by an interaction between the compression method, the compression level, and the target model’s safety boundaries.

Extractive compression frameworks (LLMLingua-2) [12] expose a systemic architectural vulnerability we termed the Extractive Vulnerability Spike. By optimizing purely for token economy, these algorithms treat low-entropy, predictable alignment cues as redundant context, systematically pruning them while retaining high-entropy adversarial directives. As

explicitly compiled in Table I and mapped in Figure 2, this structural disruption causes severe, non-monotonic jumps in the Attack Success Rate (ASR)—such as Llama-4-Scout spiking by +14.5 percentage points at $R_c = 0.7$ and Gemma-2-27B climbing by +14.0 percentage points at $R_c = 0.5$. These non-monotonic safety breaches emphasize that deploying task-agnostic extractive compressors in front of aligned LLMs poses security risks in production pipelines.

In contrast, abstractive compression (BART-large-cnn) [13] demonstrates a consistent Neutralization Effect as illustrated in Figure 3. By passing adversarial prompts through a sequence-to-sequence model biased toward factual text summarization, role-play wrappers and other adversarial disguises are stripped away. The underlying payload is reduced to its core intent, which instantly triggers the target model’s native safety filters, driving the ASR to near-zero levels across multiple architectures. Furthermore, because the abstractive compressor preserves complete grammatical and syntactic coherence, it avoids the token routing disruptions that extractive techniques introduce to Sparse Mixture-of-Experts (MoE) architectures, reducing total latency by nearly 30% while concurrently neutralizing the adversarial threat (Figure 4).

Several important avenues remain open for future work:

- 1) **Internal MoE Routing Dynamics:** A critical next step is conducting a fine-grained causal analysis of the token router within Sparse MoE architectures to definitively confirm our hypothesis regarding how syntactic fragmentation alters token-to-expert allocation and computational latency [5], [6].
- 2) **Diverse Abstractive Backbones:** Future studies should investigate if smaller, specialized, or security-tuned abstractive models can replicate BART’s neutralization benefits while minimizing the initial computational overhead of the compressor step.
- 3) **Adaptive Adversarial Exploitations:** We intend to explore adaptive, compression-aware jailbreaks to test whether adversarial frameworks can purposefully exploit both the extractive entropy pruning thresholds and the semantic floors of abstractive synthesizers to maintain their harmful intent.

ACKNOWLEDGMENT

This research was supported by the Agencia Nacional de Investigación y Desarrollo (ANID) through FONDECYT Regular Grant No. 1261266.

REFERENCES

- [1] A. Vaswani *et al.*, “Attention is all you need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 5998–6008.
- [2] L. Ouyang *et al.*, “Training language models to follow instructions with human feedback,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022, pp. 27730–27744.
- [3] R. Rafailov *et al.*, “Direct preference optimization: Your language model is secretly a reward model,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2024.
- [4] N. Shazeer *et al.*, “Outrageously large neural networks: The sparsely-gated mixture-of-experts layer,” *arXiv preprint arXiv:1701.06538*, 2017.
- [5] W. Fedus, B. Zoph, and N. Shazeer, “Switch Transformers: Scaling to trillion parameter models with simple and efficient sparsity,” *The Journal of Machine Learning Research*, vol. 23, no. 1, pp. 5232–5270, 2022.
- [6] Z. Liu *et al.*, “CompressionAttack: Exploiting prompt compression as a new attack surface in LLM-powered agents,” *arXiv preprint arXiv:2510.22963*, 2025.
- [7] K. Greshake *et al.*, “Not what you’ve signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection,” in *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security (AISec)*, 2023, pp. 79–90.
- [8] A. Wei, N. Haghtalab, and J. Steinhardt, “Jailbroken: How does LLM safety training fail?,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2024.
- [9] P. Chao *et al.*, “JailbreakBench: An open robustness benchmark for jail-breaking large language models,” in *NeurIPS Datasets and Benchmarks Track*, 2024.
- [10] A. Zou *et al.*, “Universal and transferable adversarial attacks on aligned language models,” *arXiv preprint arXiv:2307.15043*, 2023.
- [11] H. Jiang *et al.*, “LLMLingua: Compressing prompts for accelerated inference of large language models,” in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023.
- [12] H. Pan *et al.*, “LLMLingua-2: Data distillation for efficient and faithful task-agnostic prompt compression,” in *Findings of the Association for Computational Linguistics (ACL)*, 2024.
- [13] M. Lewis *et al.*, “BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension,” in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, pp. 7871–7880.
- [14] H. Inan *et al.*, “Llama Guard: Safeguarding large language models,” *arXiv preprint arXiv:2312.06674*, 2023.
- [15] P. Chao *et al.*, “Jailbreaking black box large language models in twenty queries,” *arXiv preprint arXiv:2310.08419*, 2023.
- [16] Y. Li *et al.*, “Compressing context to enhance inference efficiency of large language models,” in *Proceedings of EMNLP*, 2023.
- [17] Y. Li *et al.*, “SecurityLingua: Efficient defense of LLM jail-break attacks via security-aware prompt compression,” *arXiv preprint arXiv:2506.12707*, 2025.
- [18] AI Meta, “The Llama 4 herd of models,” *arXiv preprint*, 2026.
- [19] J. Bai *et al.*, “Qwen technical report,” *arXiv preprint arXiv:2309.16609*, 2023.
- [20] T. Gemma Team *et al.*, “Gemma 2: Improving open models through novel architecture and advanced training,” *arXiv preprint arXiv:2408.00118*, 2024.